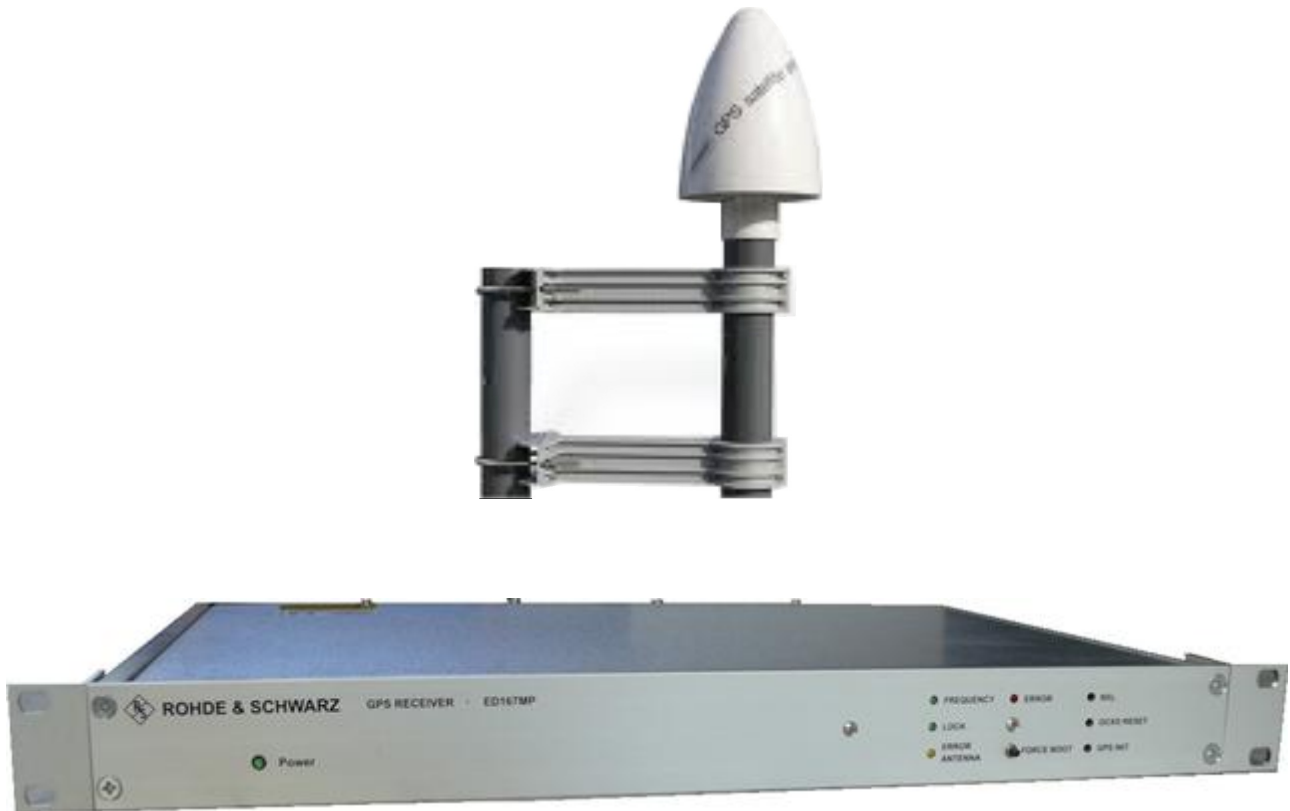




Technical Information Operating Instructions

ED167MP

Incl. Windows Software
GPSMON32E

Imprint

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General Information

The satellite receiver clock GPS167 has been designed to provide extremely precise time to its user. The clock has been developed for applications where conventional radio controlled clocks can't meet the growing requirements in precision. High precision available 24 hours a day around the whole world is the main feature of the new system which receives its information from the satellites of the Global Positioning System.

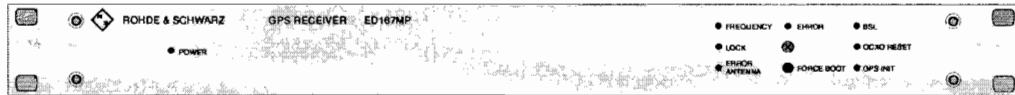
The Global Positioning System (GPS) is a satellite-based radio-positioning, navigation, and time-transfer system. It was installed by the United States Department of Defense and provides two levels of accuracy: The Standard Positioning Service (SPS) and the Precise Positioning Service (PPS). While PPS is encrypted and only available for authorized (military) users, SPS has been made available to the general public.

GPS is based on accurately measuring the propagation time of signals transmitted from satellites to the user's receiver. A nominal constellation of 21 satellites together with several active spares in six orbital planes 20000 km over ground provides a minimum of four satellites to be in view 24 hours a day at every point of the globe. Four satellites need to be received simultaneously if both receiver position (x, y, z) and receiver clock offset from GPS system time must be computed. All the satellites are monitored by control stations which determine the exact orbit parameters as well as the clock offset of the satellites' on-board atomic clocks. These parameters are uploaded to the satellites and become part of a navigation message which is retransmitted by the satellites in order to pass that information to the user's receiver.

The high precision orbit parameters of a satellite are called ephemeris parameters whereas a reduced precision subset of the ephemeris parameters is called a satellite's almanac. While ephemeris parameters must be evaluated to compute the receiver's position and clock offset, almanac parameters are used to check which satellites are in view from a given receiver position at a given time. Each satellite transmits its own set of ephemeris parameters and almanac parameters of all existing satellites.

The Modular System ED167MP

ED167MP is a set of equipment composed of a satellite controlled clock GPS167 (MQ-OCXO) together with LAN-XPT, an interface card OCTBUF for 4x10MHz / 4x PPS and a power supply unit Mean Well T-60B, all installed in a metal desktop case MULTIPAC and ready to operate. The interfaces and input/output signals provided by GPS167 are accessible via connectors in the rear and the front panel of the case. Details of the components are described below.



GPS167 in desktop case MULTIPAC (front view)

GPS167 Features

The front panel integrates 4 LED indicators, 4 push buttons and a SUB-D connector. The receiver is connected to the antenna/converter unit by a 50 ohm coaxial cable (refer to "Mounting the Antenna"). Feeding the antenna/converter occurs DC insulated via the antenna cable. Optional an antenna splitter for up to four receivers connected to one antenna is available.

GPS167 is using the "Standard Positioning Service" SPS. The altitude with its variation of $\pm 180\text{m}$ is the most inaccurate component of the position. This inaccuracy is caused by the operator (United States Department of Defense) and not by the GPS167, but it has no influence on the accuracy of the generated time. The navigation message coming in from the satellites is decoded by GPS167's microprocessor in order to track the GPS system time. Compensation of the RF signal's propagation delay is done by automatical determination of the receiver's position on the globe. A correction value computed from the satellites' navigation messages increases the accuracy of the board's oven controlled master oscillator (OCXO) and automatically compensates the OCXO's aging. The last recent value is restored from the battery buffered memory at power-up.

Time Zone and Daylight Saving

GPS system time differs from the universal time scale UTC (Universal Time Coordinated) by the number of leap seconds which have been inserted into the UTC time scale after GPS has been initiated in 1980. The current number of leap seconds is part of the navigation message supplied by the satellites, so GPS167's internal real time is based on UTC. Conversion to local time including handling of daylight saving year by year can be done by the receiver's microprocessor if the corresponding parameters are set up by the user.

Pulse and Frequency Outputs (F_SYNTH optional)

The pulse generator of GPS167 generates pulses once per second (P_SEC) and once per minute (P_MIN). Additionally, master frequencies of 10 MHz, 1 MHz and 100 kHz are derived from the OCXO. All the pulses are available with TTL level at the rear connector. The included synthesizer generates a frequency from 1/8 Hz up to 10 MHz synchronous to the internal timing frame. The phase of this output can be shifted from -360° to $+360^\circ$ for frequencies less than 10 kHz. Both frequency and phase can be setup from the front panel or using the serial port COM0. Synthesizer output is available at the rear connector as sine-wave output (F_SYNTH_SIN), with TTL level (F_SYNTH) and via an open drain output (F_SYNTH_OD). The open drain output can be used to drive an optocoupler when a low frequency is generated.

In the default mode of operation, pulse outputs and the synthesizer output are disabled until the receiver has synchronized after power-up. However, the system can be configured to enable those outputs immediately after power-up. An additional TTL output (TIME_SYN) reflects the state of synchronization. This output switches to TTL HIGH level when synchronization has been achieved and returns to TTL LOW level if not a single satellite can be received or the receiver is forced to another mode of operation by the user.

Time Capture Inputs

Two time capture inputs called User Capture 0 and 1 are provided at the rear connector (CAP0 and CAP1) to measure asynchronous time events. A falling TTL slope at one of these inputs lets the microprocessor save the current real time in its capture buffer. From the buffer, capture events are transmitted via COM1 (DCE2). The capture buffer can hold more than 500 events, so either a burst of events with intervals down to less than 1.5 msec can be recorded or a continuous stream of events at a lower rate depending on the transmission speed of COM1 can be measured. The format of the output string is ASCII, see the technical specifications at the end of this document for details. If the capture buffer is full a message "*** capture buffer full" is transmitted, if the interval between two captures is too short the warning "*** capture overrun" is being sent via COM1.

Asynchronous Serial Ports

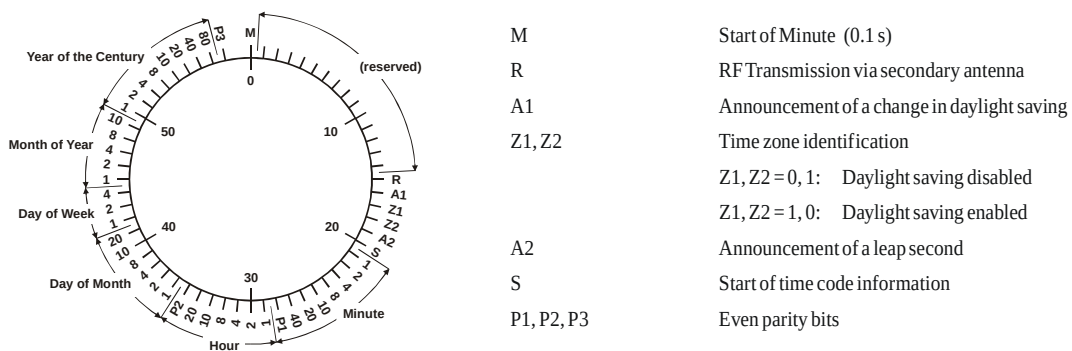
Two asynchronous serial interfaces (RS232) called COM0 and COM1 are available to the user. In the default mode of operation, the serial outputs are disabled until the receiver has synchronized after power-up. However, the system can be configured to enable those outputs immediately after power-up. Transmission speeds, framings and mode of operation can be configured separately for each interface via COM0.

COM0 is compatible with other radio remote clocks made by Meinberg. It sends Meinberg's standard time string either once per second, once per minute or on request with ASCII '?' only. COM1 can be configured for the same modes as COM0 or to transmit capture data either automatically when available or on request. The format of the output strings is ASCII, see the technical specifications at the end of this document for details. A separate document with programming instructions can be requested defining a binary data format which can be used to exchange parameters with GPS167 via COM0.

Attention: In this configuration the electrical COM0 (DCE1) can only be used for software updates (switch set to DCE1). The functions described above are standard adjustable via LAN-XPT (switch set to LAN).

DCF77 Emulation

The GPS167 satellite controlled clock generates TTL level time marks (active HIGH) which are compatible with the time marks spread by the German long wave transmitter DCF77. This long wave transmitter installed in Mainflingen near Frankfurt/Germany transmits the reference time of the Federal Republic of Germany: time of day, date of month and day of week in BCD coded second pulses. Once every minute the complete time information is transmitted. However, GPS167 generates time marks representing its local time as configured by the user, including announcement of changes in daylight saving and announcement of leap seconds. The coding scheme is given below:



Time marks start at the beginning of new second. If a binary "0" is to be transmitted, the length of the corresponding time mark is 100 msec, if a binary "1" is transmitted, the time mark has a length of 200 msec. The information on the current date and time as well as some parity and status bits can be decoded from the time marks of the 15th up to the 58th second every minute. The absence of any time mark at the 59th second of a minute signals that a new minute will begin with the next time mark. The DCF emulation output is enabled immediately after power-up.

Installation

Mounting the Antenna

The GPS satellites are not stationary but circle round the globe in a period of about 12 hours. They can only be received if no building is in the line-of-sight from the antenna to the satellite, so the antenna/converter unit must be installed in a location from which as much of the sky as possible can be seen. The best reception is given when the antenna has a free view of 8° angular elevation above horizon. If this is not possible the antenna should be installed with a mostly free view to the equator because of the satellite courses which are located between latitudes of 55° North and 55° South. If even this is not possible problems occur especially when at least four satellites for positioning have to be found.

The antenna/converter unit can be mounted on a pole with a diameter up to 60 mm or at a wall. A 50cm plastic tube, two holders for wall-mounting and clamps for pole-mounting are added to every GPS167. A standard coaxial cable with 50 ohms impedance should be used to connect the antenna/converter unit to the receiver. The maximum length of cable between antenna and receiver depends on the attenuation factor of the used coaxial cable.

Example:

Type of cable	diameter Ø [mm]	Attenuation at 100MHz [dB]/100m	max. length [m]
RG58/CU	5mm	15.9	250 ¹
RG213	10.5mm	6.9	500 ¹

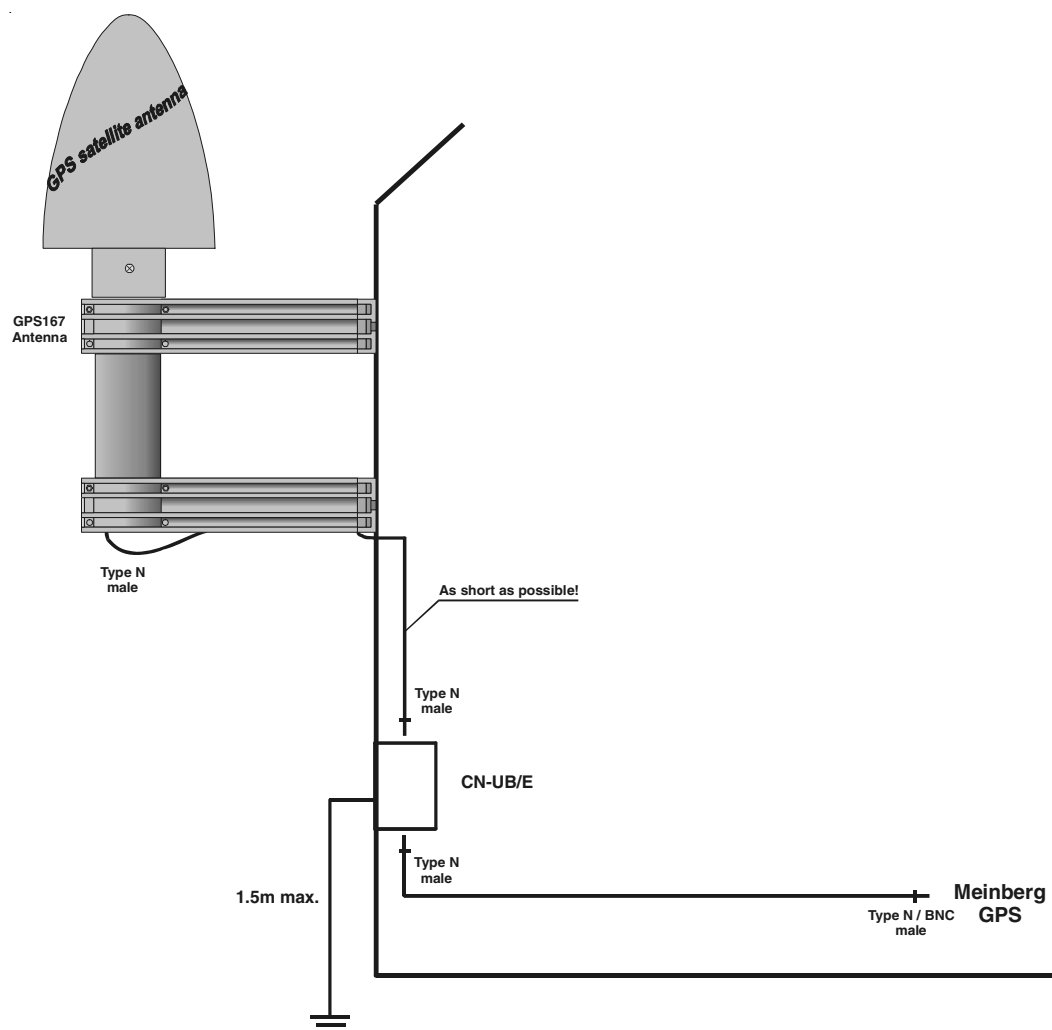
The values are typically ones; the exact ones are to find out from the data sheet of the used cable.

¹ These specifications are made for antenna/converter units produced after May, 1999. Older devices amount to 200m resp. 400m.

Up to four GPS167 receivers can be run with one antenna/converter unit by using the optional antenna splitter. The total length of one antenna line between antenna, splitter and receiver must not be longer than the max. length shown in the table above. The position of the splitter in the antenna line does not matter.

Assembly with CN-UB/E (optional)

Optionally the overvoltage protector CN-UB/E is available. Normally you directly connect the antenna converter to the system with the antenna cable.



Powering Up the System

If both the antenna and the power supply have been connected, the system is ready to operate. Between 10 seconds (with OCXO-LQ) and 3 minutes (with OCXO-MQ / HQ option) after power-up the receiver's oscillator has warmed up and operates with the specified accuracy. If the receiver finds valid almanac and ephemeris data in its battery buffered memory and the receiver's position has not changed significantly since its last operation, the receiver can find out which satellites are in view now. Only a single satellite needs to be received to synchronize and generate output pulses, therefore synchronization state can be reached within one minute (OCXO-LQ) or within 10 minutes (OCXO-MQ / HQ) after start. After 20 minutes of operation the OCXO is fully adjusted and the generated frequencies are within the specified limits.

If the receiver position has changed by some hundred kilometers since last operation, the satellites' real elevation and doppler might not match the values expected by the receiver thus forcing it to start scanning for satellites. This mode is called **Warm Boot**, and the receiver can obtain ID numbers of existing satellites from the valid almanac. When the receiver has found four satellites in view it can recalculate its position and afterwards will switch to **Normal Operation**. If the almanac has been lost due to a disconnected battery, the receiver will scan for a satellite and read in the current almanacs (every satellite periodically transmits them). This mode is called **Cold Boot**. It takes up to 12 minutes until the new almanac has been received completely and upon completion the system will switch to **Warm Boot** mode, scanning for other available satellites.

In the standard mode of operation, neither pulse and synthesizer outputs nor the serial ports will be enabled after power-up until full GPS synchronization has been achieved. However, it is possible to configure some or all of those outputs to be enabled immediately after power-up. If the system starts up in a new environment (e. g. receiver position has changed or new power supply) it can take a couple of minutes until the OCXO's output frequency has been adjusted. Until this point the frequency accuracy drops to 10^{-8} reducing the accuracy of pulses to $\pm 5\mu\text{s}$.

Control Elements in the Front Panel

FREQUENZ LED

The LED labeled "FREQUENZ" is turned on approx. 10 ... 20 minutes after powering up, when the GPS167 has reached its operating parameters and the accuracy of the output frequencies is 10^{-9} . The frequencies 2.048MHz and 10MHz are switched to the BNC connectors in the rear panel.

LOCK LED

The LOCK LED is turned on when after power-up the receiver has acquired at least four satellites and has computed its position. In normal operation the receiver position is updated continuously as long as at least four satellites can be received. In case of known and unchanged position of the receiver only one satellite is necessary to keep the internal time base synchronous to the GPS time. The LOCK LED is turned off when the accuracy of the output frequencies is reached(FREQUENZ LED).

ANTENNA LED

The LED labeled "ANTENNA" is turned on when the antenna is disconnected or defekt or the antenna cable is cut off or shorted.

ERROR LED

The ERROR LED is turned on instant after powering up until the first satellite is to be received. It is also turned on when the receiver is in the BOOTMODE (warm or cold boot) or, additional to the ANTENNA LED, when missing the antenna signal.

FORCE BOOT Key

Pushing this key lets the user force the receiver into the Boot Mode. This may be necessary when the satellite datas in the memory are too old or the receiver position has changed by some hundred kilometers since last operation. Synchronisation time may be reduced significant. If there are valid satellite datas in the memory the system starts in the WARM BOOT mode, otherwise the system changes into COLD BOOT to read new datas.

BSL Key

Whenever the on-board software must be upgraded or modified, the new firmware can be downloaded to the internal flash memory via the serial port COM0 (DCE1). There is no need to open the metal case and insert a new EPROM.

If the BSL key behind the front panel is pressed while the system is powered up, a bootstrap-loader is activated and waits for instructions from the serial port COM0 (DCE1). The new firmware can be sent to GPS167 from any standard PC with serial interface. A loader program will be shipped together with the file containing the image of the new firmware.

The contents of the program memory will not be modified until the loader program has sent the command to erase the flash memory. So if the BSL key is pressed unintentionally while the system is powered up, the firmware will not be changed accidentally. After the next power-up, the system will be ready to operate again.

OCXO RESET Key

If the key OCXO RESET behind the frontpanel is pressed the values used to discipline the OCXO are saved in an EEPROM.

GPS INIT Key

The GPS INIT key behind the front panel lets the user initialize all GPS datas, i.e. all saved satellite datas will be cleared. The system starts operating in the COLD BOOT mode and seeks for a satellite to read its actual parameters.

Firmware Updates

Whenever the on-board software must be upgraded or modified, the new firmware can be downloaded to the internal flash memory via the serial port COM0. There is no need to open the metal case and insert a new EPROM.

If the MENU key on the front panel is pressed while the system is powered up, a bootstrap-loader is activated and waits for instructions from the serial port COM0. The new firmware can be sent to GPS167 from any standard PC with serial interface. A loader program will be shipped together with the file containing the image of the new firmware.

The contents of the program memory will not be modified until the loader program has sent the command to erase the flash memory. So if the MENU key is pressed unintentionally while the system is powered up, the firmware will not be changed accidentally. After the next power-up, the system will be ready to operate again.

Skilled/Service-Personnel only: Replacing the Lithium Battery

The life time of the lithium battery on the board is at least 10 years. If the need arises to replace the battery, the following should be noted:

ATTENTION!

Danger of explosion in case of inadequate replacement of the lithium battery. Only identical batteries or batteries recommended by the manufacturer must be used for replacement. The waste battery must be disposed as proposed by the manufacturer of the battery.

Technical Specifications ED167MP

HOUSING: Metal desktop case, MULTIPAC Schroff
Front panel: 1 U / 84 HP (43.6 mm high / 426.4 mm wide)

PROTECTION
RATING: IP20

PHYSICAL
DIMENSIONS: 482,6 mm wide x 43,7 mm high x 285 mm deep

Front/Rear Panel Connectors

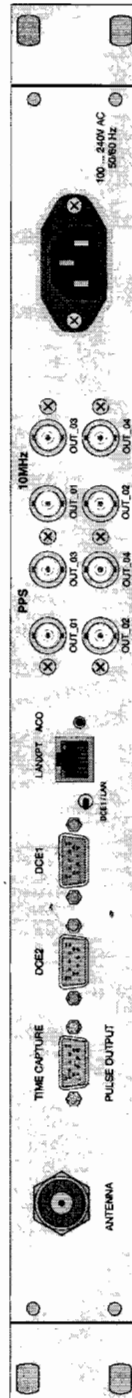
Name	Type	Signal	Cable
Rear Panel:			
DCE1 (COM0)	only Softwareupdate 9 pin SUB-D	RS-232	shielded data line
or	selectable via switch		
LANXPT	RJ45	Ethernet	CAT5 data line
DCE2 (COM1)	9 pin SUB-D	RS-232	shielded data line
Time Capture	9 pin SUB-D	TTL input	shielded line
Error Relay	(on time capt. conn)	Relay, make contact	
Antenna	Coax type N	35.4 MHz / 10 Mhz	shielded coaxial line
Power supply	power cord receptacle	90 ... 260V AC	power supply cord

Frequency Outputs

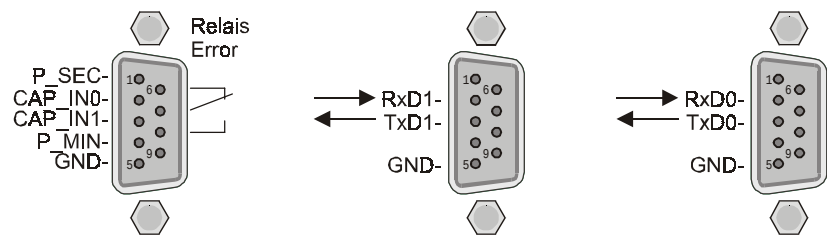
The following outputs are accessible via BNC connectors in the rear panel.

4x 10MHz	direct	2.3V _{SS}	at 50 Ohm
4x PPS	direct	2.3V _{SS}	at 50 Ohm

Rear View ED167MP



Pin Assignments of the SUB-D Connectors



Time
Capture

RS-232
COM1
DCE2

RS-232
COM0
DCE1

CE Label



This device conforms to the directive 89/336/EWG on the approximation of the laws of the Member States of the European Community relating to electromagnetic compatibility.

Technical Specifications GPS ED167MP (OCXO-MQ)

RECEIVER:	6 channel C/A code receiver with external antenna/converter unit
ANTENNA:	Antenna/converter unit with remote power supply refer to chapter "Technical Specifications GPS167 Antenna"
ANTENNA INPUT:	antenna circuit dc-insulated; dielectric strength: 1000V= Length of cable: refer to chapter "Mounting the Antenna"
TIME TO SYNCHRO- NIZATION:	one minute with known receiver position and valid almanac 12 minutes if invalid battery buffered memory
PULSE OUTPUTS:	change of second (P_SEC, TTL level) change of minute (P_MIN, TTL level)
ACCURACY OF PULSES:	after synchronization and 20 minutes of operation: TCXO HQ/OCXO LQ : better than ± 250 nsec OCXO MQ/OCXO HQ : better than ± 100 nsec better than ± 2 μ sec during the first 20 minutes of operation
FREQUENCY OUTPUTS:	10 MHz
ACCURACY OF FREQUENCY:	see Oscillator specification
FREQUENCY SYNTHESIZER:	10 MHz

SYNTHESIZER

OUTPUTS:	F_SYNTH:	TTL level	
	F_SYNTH_OD:	open drain	
		drain voltage:	< 100 V
		sink current to GND:	< 100 mA
		dissipation power at 25°C:	< 360 mW
	F_SYNTH_SIN	sine-wave	
		output voltage:	1.5 V eff.
		output impedance:	200 Ohm

TIME_SYN

OUTPUT: TTL HIGH level if synchronized

SERIAL PORTS 2 x asynchronous serial ports (RS-232)

Baud Rate: 300 up to 19200
Framing: 7N2, 7E1, 7E2, 8N1, 8N2, 8E1

default setting: COM0: 19200, 8N1
COM1: 9600, 8N1

Annotation: Even if one of the setup functions "INIT USER
PARMS" or "Resetting Factory Defaults" is
executed, the serial port parameters are
reset to default values only if invalid para
meters have been configured.

ETHERNET: 10/100 - TCP/IP

TIME CAPTURE

INPUTS: triggered on falling TTL slope
Interval of events: 1.5msec min.
Resolution: 100ns

POWER

REQUIREMENTS: 100-240 V.AC - 150 mA

PHYSICAL

DIMENSIONS: 19" module in a closed Aluminium case

FRONT PANEL: 3U / 21HP (128mm high x 107mm wide), Aluminium

REAR EDGE

CONNECTOR: according to DIN 41612, type C 64, rows a+c (male)

RF CONNECTOR: coaxial N and BNC (female)

AMBIENT

TEMPERATURE: 0 ... 50°C

HUMIDITY: 85% max.

Specifications of Internal Oscillator

	TCXO	OCXO LQ	OCXO MQ	
Kurzzeitstabilität $\tau = 1 \text{ sec}$	$2 * 10^{-9}$	$1 * 10^{-9}$	$2 * 10^{-10}$	
Freilaufgenauigkeit ein Tag	$\pm 1 * 10^{-7}$ ± 1 Hz (Note 1)	$\pm 2 * 10^{-8}$ ± 0.2 Hz (Note 1)	$\pm 1,5 * 10^{-9}$ ± 15 mHz (Note 1)	
Freilaufgenauigkeit ein Jahr	$\pm 1 * 10^{-6}$ ± 10 Hz (Note 1)	$\pm 4 * 10^{-7}$ ± 4 Hz (Note 1)	$\pm 1 * 10^{-7}$ ± 1 Hz (Note 1)	
Genauigkeit GPS-synchron 24 Std. gemittelt	$\pm 1 * 10^{-11}$	$\pm 1 * 10^{-11}$	$\pm 5 * 10^{-12}$	←
Phasenrauschen	1 Hz -60 dBc/Hz 10 Hz -90 dBc/Hz 100 Hz -120 dBc/Hz 1 kHz -130 dBc/Hz	1 Hz -60 dBc/Hz 10 Hz -90 dBc/Hz 100 Hz -120 dBc/Hz 1 kHz -130 dBc/Hz	1 Hz -75 dBc/Hz 10 Hz -110 dBc/Hz 100 Hz -130 dBc/Hz 1 kHz -140 dBc/Hz	
Stromversorgung bei 25°C Normalbetrieb Aufheizphase	+5V / 20 mA N/A	+5V / 160 mA +5V / 380 mA	+5V / 90 mA +5V / 330 mA	
Temperaturdrift im Freilauf	$\pm 1 * 10^{-6}$ (-20...70°C)	$\pm 2 * 10^{-7}$ (0...60°C)	$\pm 5 * 10^{-8}$ (-20...70°C)	

Note 1:

Die berechnete Genauigkeit in Hertz basiert auf der Normalfrequenz von 10 MHz. Zum Beispiel:

Genauigkeit des TCXO HQ (Freilauf ein Tag) ist $\pm 1 * 10^{-7} * 10 \text{ MHz} = \pm 1 \text{ Hz}$

Technical Specifications GPS167 Antenna

ANTENNA: dielectrical patch antenna, 25 x 25mm
receive frequency: 1575.42 MHz
bandwidth: 9 MHz

CONVERTER: local oscillator to converter frequency: 10 MHz
first IF frequency: 35.4 MHz

POWER

REQUIREMENTS: 12V ... 18V, @ 100mA (provided via antenna cable)

CONNECTOR: coax type N, female

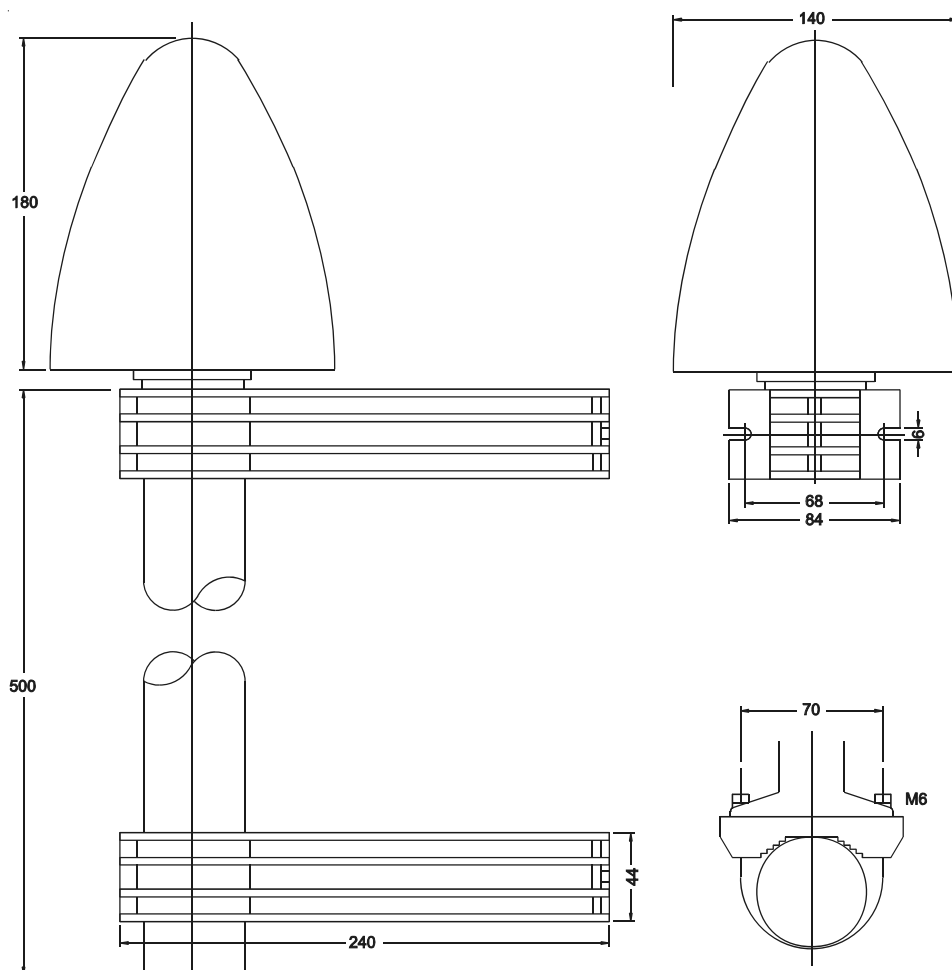
AMBIENT

TEMPERATURE: -40 ... +65°C

HOUSING: ABS plastic case for outdoor installation (IP56)

PHYSICAL

DIMENSION:



Time Strings

Format of the Meinberg Standard Time String

The Meinberg Standard Time String is a sequence of 32 ASCII characters starting with the STX (start-of-text) character and ending with the ETX (end-of-text) character. The format is:

<STX>D:dd.mm.yy;T:w;U:hh.mm.ss;uvxy<ETX>

The letters printed in *italics* are replaced by ASCII numbers whereas the other characters are part of the time string. The groups of characters as defined below:

<STX>	Start-Of-Text, ASCII Code 02h sending with one bit accuracy at change of second
dd.mm.yy	the current date: <i>dd</i> day of month (01..31) <i>mm</i> month (01..12) <i>yy</i> year of the century (00..99)
w	the day of the week (1..7, 1 = Monday)
hh.mm.ss	the current time: <i>hh</i> hours (00..23) <i>mm</i> minutes (00..59) <i>ss</i> seconds (00..59, or 60 while leap second)
uv	clock status characters (depending on clock type): <i>u</i> : ‘#’ GPS: clock is running free (without exact synchr.) PZF: time frame not synchronized DCF77: clock has not synchronized after reset ‘ ‘ (space, 20h) GPS: clock is synchronous (base accuracy is reached) PZF: time frame is synchronized DCF77: clock has synchronized after reset <i>v</i> : ‘*’ GPS: receiver has not checked its position PZF/DCF77: clock currently runs on XTAL ‘ ‘ (space, 20h) GPS: receiver has determined its position PZF/DCF77: clock is synchronized with transmitter
x	time zone indicator: ‘U’ UTC Universal Time Coordinated, formerly GMT ‘ ‘ MEZ European Standard Time, daylight saving disabled ‘S’ MESZ European Summertime, daylight saving enabled
y	announcement of discontinuity of time, enabled during last hour before discontinuity comes in effect: ‘!’ announcement of start or end of daylight saving time ‘A’ announcement of leap second insertion ‘ ‘ (space, 20h) nothing announced
<ETX>	End-Of-Text, ASCII Code 03h

Format of the Meinberg Capture String

The Meinberg Capture String is a sequence of 31 ASCII characters terminated by a CR/LF (Carriage Return/Line Feed) combination. The format is:

CH*x_tt.mm.jj_hh:mm:ss.fffffff*<CR><LF>

The letters printed in *italics* are replaced by ASCII numbers whereas the other characters are part of the time string. The groups of characters as defined below:

x 0 or 1 corresponding on the number of the capture input
– ASCII space 20h

dd.mm.yy the capture date:

dd day of month (01..31)

mm month (01..12)

yy year of the century (00..99)

hh:mm:ss.fffffff the capture time:

hh hours (00..23)

mm minutes (00..59)

ss seconds (00..59, or 60 while leap second)

fffffff fractions of second, 7 digits

<CR> Carriage Return, ASCII Code 0Dh

<LF> Line Feed, ASCII Code 0Ah

Format of the SAT Time String

The SAT Time String is a sequence of 29 ASCII characters starting with the STX (start-of-text) character and ending with the ETX (end-of-text) character. The format is:

<STX>*dd.mm.yy/w/hh:mm:ssxxxuv*<ETX>

The letters printed in *italics* are replaced by ASCII numbers whereas the other characters are part of the time string. The groups of characters as defined below:

<STX> Start-Of-Text, ASCII Code 02h
 sending with one bit accuracy at change of second

dd.mm.yy the current date:
 dd day of month (01..31)
 mm month (01..12)
 yy year of the century (00..99)

w the day of the week (1..7, 1 = Monday)

hh:mm:ss the current time:
 hh hours (00..23)
 mm minutes (00..59)
 ss seconds (00..59, or 60 while leap second)

xxxx time zone indicator:
 ‘UTC’ Universal Time Coordinated, formerly GMT
 ‘MEZ’ European Standard Time, daylight saving disabled
 ‘MESZ’ European Summertime, daylight saving enabled

u clock status characters:
 ‘#’ clock has not synchronized after reset
 ‘ ‘ (space, 20h) clock has synchronized after reset

v announcement of discontinuity of time, enabled during last hour
 before discontinuity comes in effect:
 ‘!’ announcement of start or end of daylight saving time
 ‘ ‘ (space, 20h) nothing announced

<CR> Carriage Return, ASCII Code 0Dh

<LF> Line Feed, ASCII Code 0Ah

<ETX> End-Of-Text, ASCII Code 03h

Format of the Uni Erlangen String (NTP)

The time string Uni Erlangen (NTP) of a **GPS clock** is a sequence of 66 ASCII characters starting with the STX (start-of-text) character and ending with the ETX (end-of-text) character. The format is:

<STX>*tt.mm.jj*; *w*; *hh:mm:ss*; *voo:oo*; *acdfg i;bbb.bbbbn lll.lllle hhhhm*<ETX>

The letters printed in *italics* are replaced by ASCII numbers whereas the other characters are part of the time string. The groups of characters as defined below:

<STX>	Start-Of-Text, ASCII Code 02h sending with one bit accuracy at change of second
dd.mm.yy	the current date: dd day of month (01..31) mm month (01..12) yy year of the century (00..99)
w	the day of the week (1..7, 1 = Monday)
hh.mm.ss	the current time: hh hours (00..23) mm minutes (00..59) ss seconds (00..59, or 60 while leap second)
v	sign of the offset of local timezone related to UTC
oo:oo	offset of local timezone related to UTC in hours and minutes
ac	clock status characters: a: '#' clock has not synchronized after reset ' ' (space, 20h) clock has synchronized after reset c: '*' GPS receiver has not checked its position ' ' (space, 20h) GPS receiver has determined its position
d	time zone indicator: 'S' MESZ European Summertime, daylight saving enabled ' ' MEZ European Standard Time, daylight saving disabled
f	announcement of discontinuity of time, enabled during last hour before discontinuity comes in effect: '!' announcement of start or end of daylight saving time ' ' (space, 20h) nothing announced
g	announcement of discontinuity of time, enabled during last hour before discontinuity comes in effect: 'A' announcement of leap second insertion ' ' (space, 20h) nothing announced

<i>i</i>	leap second insertion ‘L’ leap second is actually inserted (active only in 60th sec.) ‘ ‘ (space, 20h) no leap second is inserted
<i>bbb.bbbb</i>	latitude of receiver position in degrees leading signs are replaced by a space character (20h)
<i>n</i>	latitude, the following characters are possible: ‘N’ north of equator ‘S’ south d. equator
<i>lll.llll</i>	longitude of receiver position in degrees leading signs are replaced by a space character (20h)
<i>e</i>	longitude, the following characters are possible: ‘E’ east of Greenwich ‘W’ west of Greenwich
<i>hhhh</i>	altitude above sea level in meters leading signs are replaced by a space character (20h)
<ETX>	End-Of-Text, ASCII Code 03h

0Format of the NMEA 0183 String (RMC)

The NMEA String is a sequence of 65 ASCII characters starting with the '\$' character and ending with the characters CR (carriage return) and LF (line-feed). The format is:

\$GPRMC,*hhmmss.ss,A,bbbb.bb,n,llll.ll,e,0.0,0.0,ddmmyy,0.0,a*hh*<CR><LF>

The letters printed in *italics* are replaced by ASCII numbers or letters whereas the other characters are part of the time string. The groups of characters as defined below:

\$	Start character, ASCII Code 24h sending with one bit accuracy at change of second
<i>hhmmss.ss</i>	the current time: <i>hh</i> hours (00..23) <i>mm</i> minutes (00..59) <i>ss</i> seconds (00..59, or 60 while leap second) <i>ss</i> fractions of seconds (1/10 ; 1/100)
A	Status (A = time data valid) (V = time data not valid)
<i>bbbb.bb</i>	latitude of receiver position in degrees leading signs are replaced by a space character (20h)
<i>n</i>	latitude, the following characters are possible: 'N' north of equator 'S' south d. equator
<i>llll.ll</i>	longitude of receiver position in degrees leading signs are replaced by a space character (20h)
<i>e</i>	longitude, the following characters are possible: 'E' east of Greenwich 'W' west of Greenwich
<i>ddmmyy</i>	the current date: <i>dd</i> day of month (01..31) <i>mm</i> month (01..12) <i>yy</i> year of the century (00..99)
<i>a</i>	magnetic variation
<i>hh</i>	checksum (EXOR over all characters except '\$' and '*')
<CR>	Carriage Return, ASCII Code 0Dh
<LF>	Line Feed, ASCII Code 0Ah

Format of the ABB SPA Time String

The ABB SPA Time String is a sequence of 32 ASCII characters starting with the characters ">900WD" and ending with the <CR> (Carriage Return) character. The format is:

>900WD;*yy-mm-tt_hh.mm:ss.fff*:*cc*<CR>

The letters printed in *italics* are replaced by ASCII numbers whereas the other characters are part of the time string. The groups of characters as defined below:

<i>yy-mm-tt</i>	the current date:	
<i>yy</i>	year of the century	(00..99)
<i>mm</i>	month	(01..12)
<i>dd</i>	day of month	(01..31)
—	Space (ASCII code 20h)	
<i>hh.mm:ss.fff</i>	the current time:	
<i>hh</i>	hours	(00..23)
<i>mm</i>	minutes	(00..59)
<i>ss</i>	seconds	(00..59, or 60 while leap second)
<i>fff</i>	milliseconds	(000..999)
<i>cc</i>	Check sum. EXCLUSIVE-OR result of the previous characters, displayed as a HEX byte (2 ASCII characters 0..9 or A..F)	
<CR>	Carriage Return, ASCII Code 0Dh	

Format of the Computime Time String

The Computime time string is a sequence of 24 ASCII characters starting with the T character and ending with the LF (line feed, ASCII Code 0Ah) character. The format is:

T:yy:mm:dd:ww:hh:mm:ss<CR><LF>

The letters printed in *italics* are replaced by ASCII numbers whereas the other characters are part of the time string. The groups of characters as defined below:

T	Start character sending with one bit accuracy at change of second
yy:mm:dd	the current date: yy year of the century (00..99) mm month (01..12) dd day of month (01..31)
ww	the day of the week (01..07, 01 = monday)
hh:mm:ss	the current time: hh hours (00..23) mm minutes (00..59) ss seconds (00..59, or 60 while leap second)
<CR>	Carriage Return, ASCII Code 0Dh
<LF>	Line Feed, ASCII Code 0Ah

Signal Description GPS167 V7.x

Name	Pin	Function
GND	32a+c	Ground
VCC in (+5V)	1a+c	+5V supply
VCC in (+12V)	2a+c	+12V supply
VCC in (+5V)	3a+c	+5 V supply (TCXO / OCXO)
P_SEC out	6c	Pulse when second changes, TTL level, active high, length 200 msec
P_MIN out	8c	Pulse when minute changes, TTL level, active high, length 200 msec
/RESET in/out	9c	RESET signal, Open Drain pulled up to +5V
Prog. Pulse out	10c-12c	programmable pulse, TTL level
100 kHz out	10a	100 kHz frequency output, TTL level
1 MHz out	11a	1 MHz frequency output, TTL level
10 MHz out	12a	10 MHz frequency output, TTL level
TIMECODE DC	13a	Timecode unmod. out
TIMECODE AM	14a	Timecode mod. out 3Vpp
DCF_MARK out	17c	DCF77 compatible second marks, TTL level active high, length 100/200 msec
TIME_SYN	19c	TTL output, HIGH level if synchronization has been achieved, LOW level after reset or in case of serious errors (e.g. antenna faulty)
F_SYNTH	21c	Synthesizer output, TTL-Pegel
F_SYNTH_OD	22c	Synthesizer output, Open Drain, max sink current to GND: 150mA
F_SYNTH_SIN	23c	Synthesizer output, sine-wave 1.5 V eff.
CAPx	27c, 28c	Time capture inputs (TTL), capture on falling slope
COMx TxD out		COMx RS-232 transmit data output
COMx RxD in		COMx RS-232 receive data input
SDA, SCL, SCL_EN (reserved)		internal serial control bus, for extension boards reserved, do not connect

Rear Connector Pin Assignments GPS167 V7.x

	a	c
1	VCC in (+5V)	VCC in (+5V)
2	VCC in (+12V)	VCC in (+12V)
3	VDD in (TCXO/OCXO)	VDD in (TCXO/OCXO)
4	(reserved, FreqAdjust out)	
5		
6		P_SEC out
7		
8	(reserved, 10 MHz in)	P_MIN out
9		/RESET in/out
10	100 kHz out	ProgPulse0 out
11	1 MHz out	ProgPulse1 out
12	10 MHz out	ProgPulse2 out
13	TIME CODE DC out	SCL
14	TIME CODE AM out	SCL_EN
15	COM2 RxD in	SDA
16	COM2 TxD out	(reserved, P3.2)
17	COM3 RxD in	DCF_MARK out
18	COM3 TxD out	(reserved, Vref/TxD2 TTL)
19	GND	TIME_SYN out
20	GND	(reserved, P2.3)
21	GND	F_SYNTH out
22	GND	F_SYNTH_OD out
23	GND	F_SYNTH_SIN out
24	GND	COM1 TxD out
25	GND	
26	GND	COM0 TxD out
27	GND	CAP1 in
28	GND	CAP0 in
29	GND	COM1 RxD in
30	GND	COM0 RxD in
31	GND	GND
32	GND	GND

male connector according to DIN 41612, type C 64, rows a + c

Technical Specifications Power Supply Unit

INPUT:	85 ... 264V AC, 47 ... 63Hz, 1A/230V, 2A/115V		
FUSE:	electronic		
CURRENT LIMITING:	105 - 150% I _{out nom}		
OUTPUTS:	V _{out} 1:	5.05V / 5A	
	V _{out} 2:	+12V / 2.5A	
	V _{out} 3:	-12V / 0.5A	
TOTAL LOAD:	max. 61Watt		
CONNECTORS:	screw terminal		
HOUSING:	metal housing : 159mm x 97mm x 38mm		
AMBIENT TEMPERATURE:	-10°C ... +60°C		
HUMIDITY:	90% max.		

Control LEDs

POWER:	LED green, output voltage OK.
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LAN-XPT Networking Functions:

The LANXPT expansion module transfers the first serial port (COM0) of a Meinberg GPS based radio clock into a 10baseT/100baseT Ethernet interface, adding functionality that allows a user to query status informations by using the TCP/IP protocol and the Meinberg application GPSMON32 or a third-party SNMP management software. It is also possible to connect to the radio clock via network, reading the time and date telegrams generated by the clock. If configured, the module automatically queries the clock periodically and generates alarm messages (SNMP traps), which are sent to the SNMP management software. Additionally a configured syslog server can receive those alarm messages and record them for later reference.

Controls:

LEDs

Two orange LEDs are included in the RJ45 network connector of your LANXPT module. The left one is indicating the duplex mode of the network connection (LED on = full duplex, LED off = half duplex), the right one blinks when there is network activity.

Additionally there are two LEDs to the left of the network connector. The green diode is indicating that the module is in operation mode, the orange LED is on during ACO (Access Control Override) mode, see below for further information.

ACO-Button (Access Control Override)

The small button to the right of the network connector can be used if the LANXPT password has been forgotten. It is only operational when no network connection is active. When pressed, the LANXPT password of the module is reset to "nothing" (meaning an empty password, just press ENTER) for 30 seconds, afterwards the password switches back to its old value. During this ACO-mode the orange LED is switched ON.

You can access the telnet setup and open up a GPSMON32 connection with this empty password, as long as the orange ACO-LED is on. Please note: If you open up a GPSMON32 connection while in this mode, the "empty" password is active during the

whole session, it will be reset to the old value only after the connection has been closed (if you change it, the new one is activated, of course).

You can enter a new password by using the telnet setup (menu option 90) or the GPSPMON32 software. If you use telnet setup, please do not forget to save and reboot (menu option 9) immediately after you entered the new password.

Configuration:

Initial configuration of the network parameters

In order to be able to connect to the module over the network, three parameters have to be set up correctly. It is advisable to contact your local network administrator before installing the unit in order to inform her/him about the new network device and eventually ask for the necessary parameters. This may avoid severe network problems, e.g. caused by double IP addresses.

IP Address

The unique IP address of your LANXPT module. It must be selected from within a range which is valid for the local network. Such an IP address normally has to be defined by the network administrator.

If the IP address is set to "000.000.000.000", then the module will try to request an IP address from a DHCP server. If no DHCP server is reachable, it will try to assign itself an address from the so-called AutoIP-range (169.254.0.1 - 169.254.255.1). This can be avoided by assigning an IP address "000.000.001.000", representing the "DHCP but no AutoIP" choice. AutoIP is also not used if a DHCP server is reachable but denies to assign an IP address to the module. In this case, the unit continuously sends DHCP requests until it gets an address. If AutoIP is disabled (by using "0.0.1.0" as the IP address), your LANXPT module will continuously try to find a DHCP server accepting its request, too. After the unit received a valid DHCP assigned IP address, it will restart using this address.

Netmask

The netmask defines whether a destination IP address belongs to the local network and is needed for routing purposes. Detailed information about this topic can be found in TCP/IP protocol literature. You may also ask your network administrator or search information on the internet.

Default Gateway

All IP addresses the LANXPT module communicates with will be classified as local or remote partners, depending on the unit's own IP address and the netmask. If a destination address is found to be remote ("not reachable directly"), all communication traffic will be routed over the default gateway, that is the host configured as the "gateway" in the module configuration.

These three network parameters can be entered by using the radio clocks display and front panel buttons (if supported by the clock's firmware). You will find a "SETUP IP CFG SETTINGS" menu in the setup of these models shortly after the clock detected the LANXPT module at COM0. This happens a few seconds after the

LANXPT module has been connected to the clock's COM0 respectively directly after the radio clock has been powered up.

For all models without a display and/or front panel buttons and all of the GPS radio clocks with a firmware version not supporting the LANXPT module, the configuration must be done with the GPSPMON32 application. A detailed description for these setup procedures can be found in the documentation of the GPSPMON32 software. Please note that you must use version 2.x or later of GPSPMON32, earlier versions do not include network support.

If the LANXPT module is already reachable via network, you can setup the network parameters by using the telnet setup, which is described in detail in the next section of this manual.

Extended configuration with telnet setup

Once the network parameters of the LANXPT module are set up correctly, you can configure the unit by using a telnet program. You have to know the IP address of your module and then connect to port 9999 at this address with your telnet client application (see the documentation of your telnet software on how to initiate a connection). After the connection has been established, you are asked for the LANXPT password (default is "meinberg").

Please be aware of the fact that a telnet connection always is unencrypted (even the password is sent as plain text during the login sequence), so you should make sure that no sniffing is possible when you are accessing your LANXPT module by telnet. This can be achieved by using a crosslink-cable and a direct connection between the module and, for example, a laptop.

The changes to the setup you are applying during the telnet session will not be effective until you leave the setup dialogue by choosing option 9 (save and exit). Only then the changed parameters are stored permanently, afterwards the module will be rebooted (takes about 10 to 30 seconds) to activate the changes.

When you see the welcome message of the module after entering the correct password, please press Enter to go into the setup menu:

```
*** Meinberg LANXPT Setup V1.02 ***  
MAC address 00204A80FBC3  
Software version 01.0b1 (040607) MBG_LANXPT  
LANXPT Password:*****
```

Press Enter to go into Setup Mode

After pressing Enter, the important parameters are displayed and the menu appears:

*** basic parameters

Hardware: Ethernet TPI

IP addr 172.16.3.202, no gateway set, netmask 255.255.255.000

Clock port (1)

Baudrate 19200, I/F Mode 4C, Flow 00

Source Port (10001)

SNMP Configuration:

SNMP community name for read: public

SNMP community name for write: private

Trap IP addresses:

1: " not set "

2: " not set "

3: " not set "

SYSLOG Configuration:

SYSLOG is enabled

SYSLOG server : 172.016.003.002

Change Setup:

1 Network configuration

2 Clock port configuration

3 SNMP configuration

4 SYSLOG configuration

7 factory defaults

8 exit without save (no reboot)

9 save and exit

90 Change password

Your choice ?

You may now choose a menu option by entering the corresponding number (1,2,3,7,8,9 or 90) and confirm your choice by pressing ENTER.

The available menu options are described below.

8.4.1 Menu option 1 Network configuration

The setup of the network parameters can be changed by using this menu option:

```
IP Address : (172) .(016) .(003) .(202)
Set Gateway IP Address (Y) Y
Gateway IP Address : (172) .(016) .(003) .(002)
Netmask: Number of Bits for Host Part (0=default) (8)
```

Note: In the telnet setup, IP addresses are entered as four single decimal digits (0-255). The setup always shows the actual value in brackets and waits for an input:

```
IP Address : (172) _
```

"_" represents the cursor. Enter a new value and press Enter. If you press Enter without entering a new value, the old value will be used. After pressing Enter, the next part of the IP address appears:

```
IP Address : (172) 192 .(016) _
```

Now you can enter the second part of the IP address and confirm with Enter. Then enter the third and fourth part the same way. The line could look like this afterwards:

```
IP Address : (172) 192 .(016) 168 .(003) 100 .(202) 11
```

So in this example you changed the IP address from 172.16.3.202 to 192.168.100.11 !

When the setup asks for the netmask, you have to enter the number of set bits forming your desired netmask. If you want to use a netmask like 255.255.255.0, this is represented by 8 bits for the host part (the last 8 bits are zero).

8.4.2 Menu option 2 Clock port configuration (LAN-XPT / SCU-XPT)

These parameters are defining the way the XPT module communicates with the radio clock it is attached to (SCU-XPT and LAN-XPT models). If you change anything here, it is very likely that the communication with the clock will be broken. Additionally there may arise problems for GPSMON32 to talk to the module in a proper way. Therefore we recommend not to alter the default settings in any way:

```
Baudrate (19200)
I/F Mode (4C) ?
Flow (00) ?
Source Port (10001)
```

If you changed something here accidentally, just re-enter the above shown default values (in brackets) manually or choose the menu option 7 (factory defaults) in order to reset the whole configuration (this changes almost everything, so it is safer to reenter the shown settings manually).

8.4.3 Menu option 3 SNMP Configuration

The parameters configurable with menu option 3 are related to SNMP:

SNMP community name for read (public):

The SNMP community is some kind of rudimentary access control mechanism. Each SNMP request is provided with a community string, which is checked by the modules SNMP subsystem. If the string corresponds to this "SNMP community for read" parameter, read-only access will be granted. Attention: Because the community string is not a very secure protection (it is transferred in plain text over the network), the LANXPT module additionally checks if the sender of the SNMP request is defined as a SNMP trap receiver (see below). If not, the access is denied.

SNMP community name for write (private):

If an incoming SNMP request provides this community string, the LANXPT SNMP subsystem grants read-write access. Because there is no SNMP write-supported variable available at the moment, this parameter can be used to define a second SNMP community allowed to query the unit.

SNMP community name for traps (pubtrap):

This community is used for sending the SNMP traps to the trap receivers.

Enter IP addresses for SNMP traps:

1: (172) .(016) .(003) .(002)
2: (172) .(016) .(003) .(045)
3: (000) .(000) .(000) .(000)

Here you may define up to three SNMP trap receivers. These IP addresses are also used as a security option, because your LANXPT only replies to SNMP requests coming from one of these three IP addresses. The input of IP addresses is described in detail above, please refer to menu option 1 !

The description of all supported and provided SNMP variables can be found in the "SNMP reference" section in this manual.

8.4.4 Menu option 4 Syslog Configuration

This menu option allows you to specify a syslog server to which the XPT module sends alarm and status messages by using UDP and port 514. The syslog server then handles those messages according to its own configuration. It may just record them in its own system log or invokes other actions. Please note that you will have to alter the configuration of your syslog server software to tell it that it is allowed to receive these messages from the XPT module and how to handle them. Please check the documentation of your syslog server software to find out how to do this in a proper way.

The dialogue for changing the syslog server entry looks like this:

******* SYSLOG Configuration *******

Use SYSLOG logging? (Y) Y

Enter IP address for SYSLOG server:(172) .(016) .(003) .(002)

8.4.5 Menu option 7 Factory Defaults

With this option you can reset your module configuration to the default settings. Please note that the network parameters are NOT changed. If you ever set up your unit the wrong way, you may restore the original settings by using this option.

8.4.6 Menu option 8 Exit without save

If you leave the setup by choosing this menu option, all changes you applied to the setup during the actual telnet session will not be stored permanently. The setup remains in the state until you reboot/power cycle it.

8.4.7 Menu option 9 Save and exit

All settings will be stored in the non-volatile memory of the module, afterwards it will reboot in order to activate the changes. Only the LANXPT module will reboot, the radio clock is not affected by this in any way! The boot process takes up to 30 seconds, afterwards the unit is running with the new settings and may be contacted again over the network.

8.4.8 Menu option 90 Change password

The XPT password that is used for authentication (telnet/GPSMON32) and for encryption can be changed by choosing this menu option. Please note that any currently active GPSMON32 connection will be interrupted due to the changing encryption keys.

Enter new XPT Password :*****

Confirm new XPT Password:*****

Password changed. The new password is used immediately, but you have to save your settings in order to use it permanently. However, every active session (if any) has been interrupted due to the password change.

Press any key to return to main menu:

As noted above, because the entered new password (you have to enter it twice to avoid a typo as it is not shown during input) is activated immediately, an open GPSPMON32 connection is interrupted due to the password change and the corresponding crypto key change. After you changed the password, you therefore have to reconnect.

SNMP reference

The SNMP support of the LANXPT module enables you to monitor the status and health of your Meinberg GPS radio clock by using a SNMP management software. This allows you to reduce reaction times after a failure, for example a broken antenna. Your SNMP management software can query the LANXPT module for a number of status indicators and values, additionally the module itself sends SNMP traps to up to three management systems when a change of the status conditions is recognized. See below for a description of all possible variables and trap generating events.

SNMP variables are represented by so-called object identifiers (OIDs). An OID is a sequence of numbers building a hierarchical structure (tree-like, with root, branches and leaves). A detailed description can be found at the net-snmp website (www.net-snmp.org), that is an open source project dealing with the creation of SNMP software.

An OID looks like this: "1.3.6.1.4.1.5597.10.2.2"

Meinberg equipment is arranged under the "1.3.6.1.4.1.5597" branch, the 5597 is the unique Meinberg Radio Clocks official enterprise ID. All variables dealing with Meinberg GPS radio clocks can be found under the "10" branch, so all OIDs related to the GPS versions of the LANXPT module are beginning with "1.3.6.1.4.1.5597.10".

Here is a schematic representation of all available mbgGPS variables:

```
+--mbgGPS(10)
|
|   +--mbgGPSRefclock(2)
|   |   |
|   |   |   +- -R- String      mbgGPSRefClockType(1)
|   |   |   +- -R- Integer32  mbgGPSRefClockTypeVal(2)
|   |   |   +- -R- String      mbgGPSRefClockMode(3)
|   |   |   +- -R- Integer32  mbgGPSRefClockModeVal(4)
|   |   |   +- -R- String      mbgGPSRefGpsState(5)
|   |   |   +- -R- Integer32  mbgGPSRefGpsStateVal(6)
|   |   |   +- -R- String      mbgGPSRefGpsPosition(7)
|   |   |   +- -R- String      mbgGPSRefGpsSatellites(8)
|   |   |   +- -R- Integer32  mbgGPSRefGpsSatellitesGood(9)
|   |   |   +- -R- Integer32  mbgGPSRefGpsSatellitesInView(10)
|   |   |
|   |   +--mbgGPSTraps(3)
|   |   |   |
|   |   |   |   +-mbgGPSTrapColdBoot(1)
|   |   |   |   +-mbgGPSTrapWarmBoot(2)
|   |   |   |   +-mbgGPSNavSolved(3)
|   |   |   |   +-mbgGPSTrapReceiverNotResponding(4)
|   |   |   |   +-mbgGPSTrapReceiverNotSync(5)
|   |   |   |   +-mbgGPSTrapAntennaFaulty(6)
|   |   |   |   +-mbgGPSTrapAntennaReconnect(7)
|   |   |   |   +-mbgGPSTrapLANXPTBoot(8)
|   |   |   |   +-mbgGPSTrapLeapSecondAnnounced(9)
|   |   |   |   +-mbgGPSTrapTestNotification(99)
```

Below, you find the description of all status variables of your GPS clock (mbgGPS-Refclock, OID 1.3.6.1.4.1.5597.10.2) and all possible GPS alarm messages (mbgGPSTraps, OID 1.3.6.1.4.1.5597.10.3) used by the LANXPT module.

SNMP variables

All supported SNMP variables provided by the LANXPT module are updated periodically (every 1-2 seconds) by querying the attached radio clock. This automatic data update process is suspended, if a network session is initiated by a GPSPMON32 client or any other software connects to the LANXPT module via network. Because the unit is able to analyse the communication between the radio clock and a GPSPMON32 client, the status variables are still continuously updated during a GPSPMON32 connection. The status data remains unchanged, if any other software uses the LANXPT port for accessing the radio clock, meaning no traps are generated during such a session and no status update takes place. As soon as the network session is closed, the module continues with its automatic data update queries.

The following variables are available:

mbgGPSRefClockType (OID 1.3.6.1.4.1.5597.10.2.1)

Contains the model name of the radio clock attached to the LANXPT module as a string (example: "GPS167").

mbgGPSRefClockTypeVal (OID 1.3.6.1.4.1.5597.10.2.2)

Contains the model type of the radio clock attached to the LANXPT module as an integer value (example: 2).

mbgGPSRefClockMode (OID 1.3.6.1.4.1.5597.10.2.3)

Is a string containing the actual operation mode of the radio clock (examples: "NORMAL OPERATION", "COLD BOOT", "WARM BOOT").

mbgGPSRefClockModeVal (OID 1.3.6.1.4.1.5597.10.2.4)

The actual operation mode represented as an integer value (example: 2).

mbgGPSRefGpsState (OID 1.3.6.1.4.1.5597.10.2.5)

In this variable the synchronisation state of the clock is stored as a string ("SYNC", "NOT SYNC" or "N/A" if it could not yet be detected by the module).

mbgGPSRefGpsStateVal (OID 1.3.6.1.4.1.5597.10.2.6)

The synch state of the attached clock as an integer value (0=not sync, 1=sync, -1=unknown).

mbgGPSRefGpsPosition(OID 1.3.6.1.4.1.5597.10.2.7)

The position of the receiver, this is not supported by all GPS clocks (example: "51°59'00 N, 9°13'33 E, 183m").

mbgGPSRefGpsSatellites (OID 1.3.6.1.4.1.5597.10.2.8)

This variable contains a string showing the number of detected satellites ("in view") and the number of satellites suitable for synchronisation ("good"). The string format is "GPS-Satellites: GG / II", where "GG" represents the number of good satellites and "II" stands for the number of satellites in view.

mbgGPSRefGpsSatellitesGood (OID 1.3.6.1.4.1.5597.10.2.9)

The number of satellites suitable for synchronisation ("good"), represented by an integer value (example: 8)

mbgGPSRefGpsSatellitesInView (OID 1.3.6.1.4.1.5597.10.2.10)

The number of detected satellites ("in view") as an integer value (example: 9).

SNMP traps (alarm messages)

The LANXPT module constantly monitors the state of the attached radio clock and sends out SNMP traps whenever specific events occur. Up to three trap receivers may be configured, on which a SNMP management software could take special actions upon reception of such a trap. For example, service personnel could be alarmed automatically or a simple log entry could be written.

If a customer specific software connects to the LANXPT over the network, the periodic updates of the SNMP variables are suspended until this network connection is closed. Because the Meinberg GPSSMON32 application itself queries the clock on a regular basis, the LANXPT module can update its status information by analysing the traffic between the GPSSMON32 software (requesting status information) and the radio clock (replying with status information). So the monitoring procedure is not disturbed by using GPSSMON32 to connect to the clock over the network.

The following traps are sent:

mbgGPSTrapColdBoot (OID 1.3.6.1.4.1.5597.10.3.1)

Whenever the GPS clock switches to cold boot mode, this trap is sent.

mbgGPSTrapWarmBoot (OID 1.3.6.1.4.1.5597.10.3.2)

If the clock enters warm boot mode, the module generates this trap.

mbgGPSNavSolved (OID 1.3.6.1.4.1.5597.10.3.3)

This trap is sent whenever the clock managed to calculate its position and synchronized to the GPS time.

mbgGPSTrapReceiverNotResponding (OID 1.3.6.1.4.1.5597.10.3.4)

If the LANXPT module did not manage to talk to the clock for a while, this trap is generated.

mbgGPSTrapReceiverNotSync (OID 1.3.6.1.4.1.5597.10.3.5)

If the receiver loses synchronisation, this trap is sent. Often this happens in conjunction with another trap event (Antenna faulty, for example).

mbgGPSTrapAntennaFaulty (OID 1.3.6.1.4.1.5597.10.3.6)

If the connection to the antenna is broken, the module sends this trap.

mbgGPSTrapAntennaReconnect (OID 1.3.6.1.4.1.5597.10.3.7)

This trap indicates that the clock re-detected a connection to the antenna or that the antenna has been repaired.

mbgGPSTrapLANXPTBoot (OID 1.3.6.1.4.1.5597.10.3.8)

Every time after the LANXPT module restarts and finished its boot process, it sends this trap. This happens after power up and after a reboot triggered by a change of configuration.

mbgGPSTrapLeapSecondAnnounced (OID 1.3.6.1.4.1.5597.10.3.9)

If a leap second has been announced by the GPS, the modul sends this trap.

mbgGPSTrapTestNotification(99)

This trap is for test purposes only and not used at present.

The Meinberg MIB

SNMP Management software needs a description of the provided SNMP variables and traps for a unit it wants to monitor / manage. This description is stored in textfiles in a special notation. All of those textfiles together build the MIB (Management Information Base). The part of the MIB containing the information about your Meinberg equipment is called Meinberg MIB and consists of a group of those textfiles. You should find a copy of those files on the installation media shipped together with your radio clock or single LANXPT module. If not, please visit our website in order to download the latest version of the Meinberg MIB files for your LANXPT module.

Output Driver card OCTBUFV30

Functioning

The expansion board OCTBUF drives up to eight outputs with pulses or frequencies via 50Ohm. The device is provided with a TTL level input.

Technical Specifications OCTBUF

POWER

REQUIREMENTS : +5V @300mA

INPUT : TTL (e.g.: P_SEC or 10MHz out from the GPS167)

OUTPUT : 8 x approx. 2.5V_{ss} at 50Ohm

PHYSICAL

DIMENSION : 100mm x 36mm

AMBIENT

TEMPERATURE : 0 ... 50° C

HUMIDITY : 85% max.

12 The program GPSMON32



The program GPSMON32 can be used to monitor and programm all essential functions of Meinberg GPS- Receivers. The Software is executable under Windows 7, Windows Vista, Win9X, Win2000, WinXP and WinNT. To install GPSMON32 just run **setup.exe** from the included USB flash drive and follow the instructions of the setup program.

Program and clock can communicate either via serial link or via TCP/IP connection if the clock is prepared for (LANXPT or SCU-XPT board). The mode to be used can be selected in menu *"Connection -> Settings"* by the checkboxes serial and network.

12.1 Serial Connection

To obtain a connection between you PC and the GPS receiver, connect the receivers COM0 port to a free serial port of your PC. The PCs comport used by the program GPSMON32 can be selected in submenu "PC-Comport" in menu "Connection".

Also transfer rate and framing used by the program are selected in this menu. Communication between the clock and the PC comes about, only if the GPS serial port is configured in the same way as the PCs comport. You can enforce an access, if the GPS serial port is not configured with appropriate parameters for communication. Select the menu item "Enforce Connection" in menu "Connection" and click "Start" in the apperaring window. Some firmware versions of the GPS receiver do not support this way of setting up a connection. If "Enforce Connection" doesn't succeed apparently, please change the serial port parameter of GPS COM0 manually to the PCs parameters.

12.2 Network Connection

Settings needed for a network connection can be done in menu "Connection->Settings".

To set up a network connection from clock program GPSMON32, the mode "network" must be selected field "mode". Further the TCP/IP-Address must be entered in field "IP-Address". If the IP-Address is unknown, user can let the program search for available clocks in the local network by clicking the "Find" button. A new connection can be set up by clicking to one of the displayed addresses.

Access to radioclocks by network is always protected by a Password.

The online help function of GPSMON32 provides detailed information on setting up a TCP/IP connection.

12.3 Online Help

The online help can be started by clicking the menu item "Help" in menu Help. In every program window a direct access to a related help topic can be obtained by pressing F1. The help language can be selected by clicking the menu items German/English in the Help Menu

